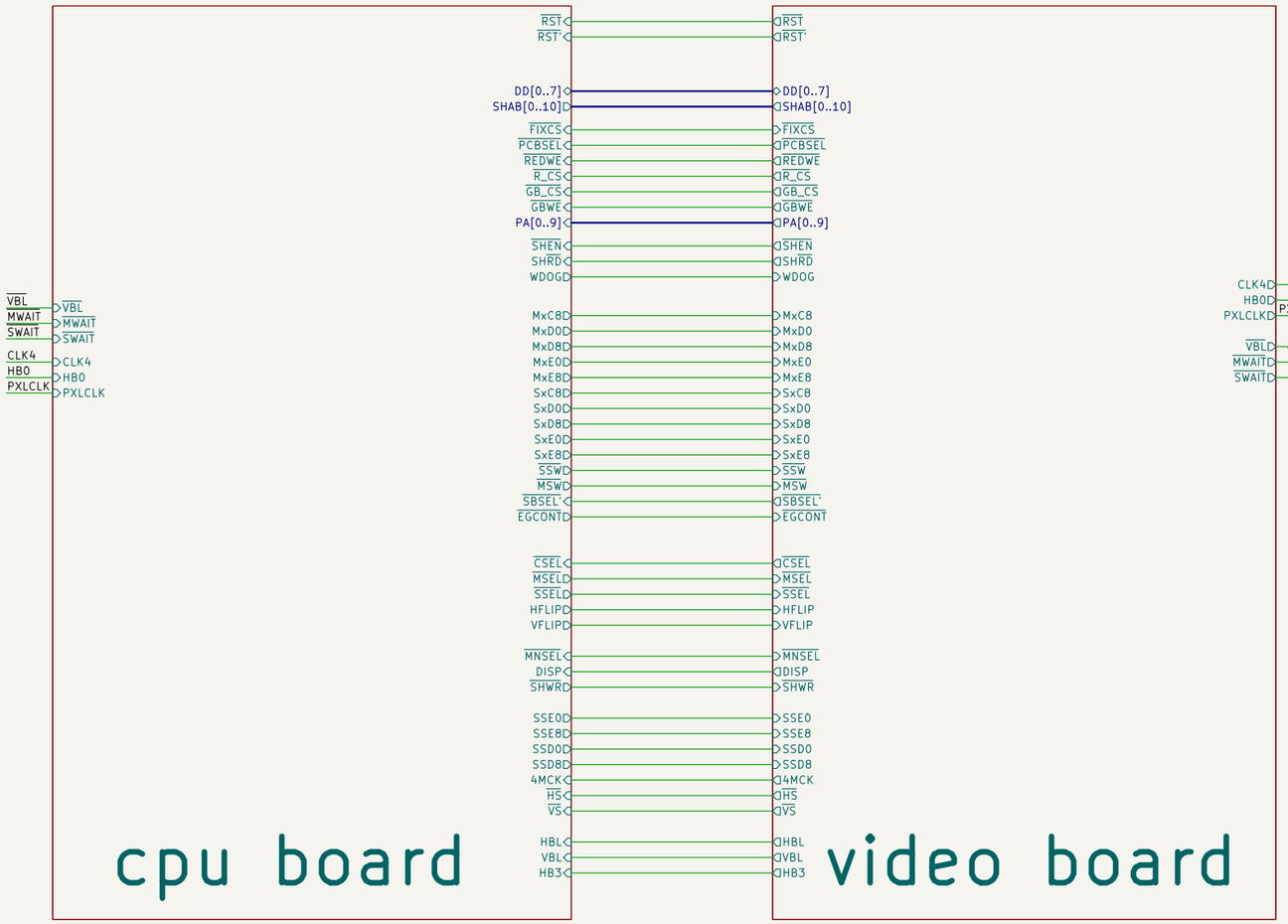


tehkanwch bootleg board

- original PCB has through-hole vias whereas the bootleg has filled vias
- original uses three custom packages from Mitsubishi with logic inside.
  These were replaced by a small daughter board on the bootleg, serving the same functionality
- different edge connectors
- debug edge connector on video board not present on bootleg
- cabinet input logic partly missing and rewired on the bootleg to replace the trackball interface for a joystick one
- different power amplifier for sound without heat sink on the bootleg
- slightly different frequencies for crystal oscillators (cost reduction?)
- device reference markers (silkscreen) on bootleg have reversed order on bootleg

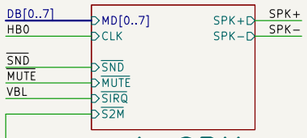
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Sheet: /
File: tehkanwch.kicad_sch

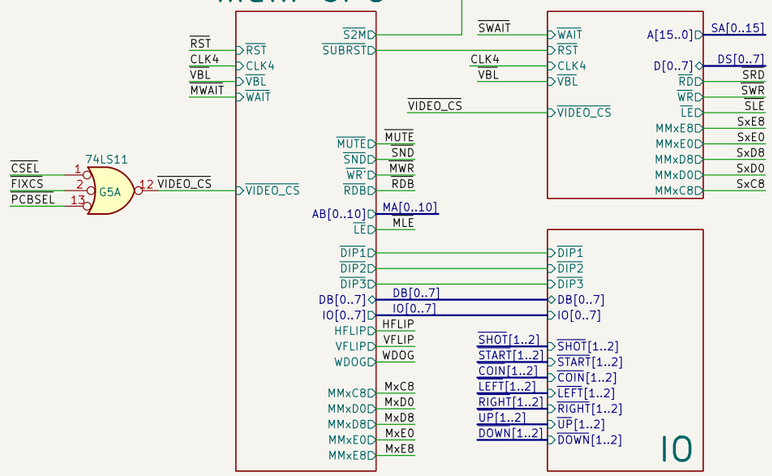
Title: TEHKAN WORLD CUP (tehkanwch bootleg)

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 10.0.4		Id: 1/26

sound



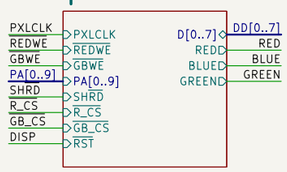
main CPU



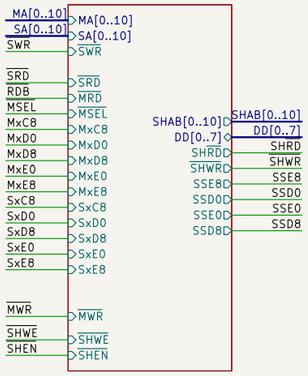
sub CPU

IO

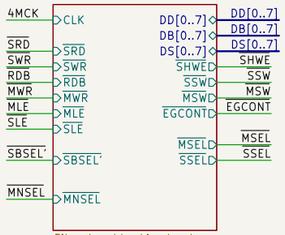
palette



shared

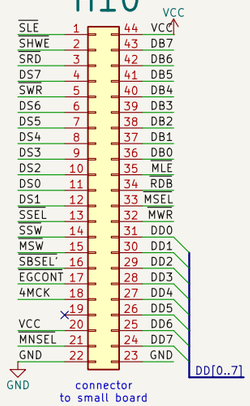


daughter

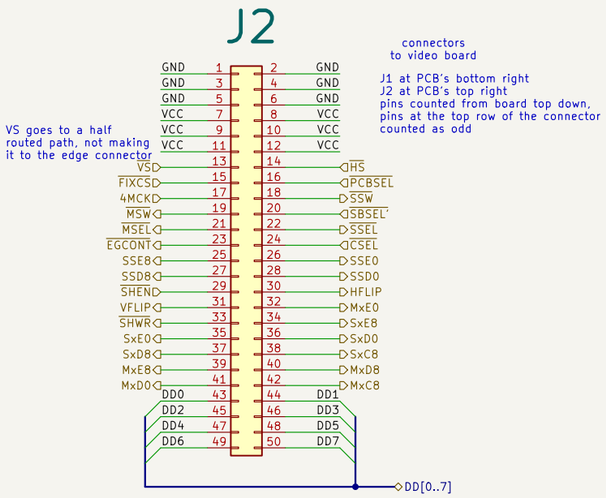


File: daughter.kicad_sch

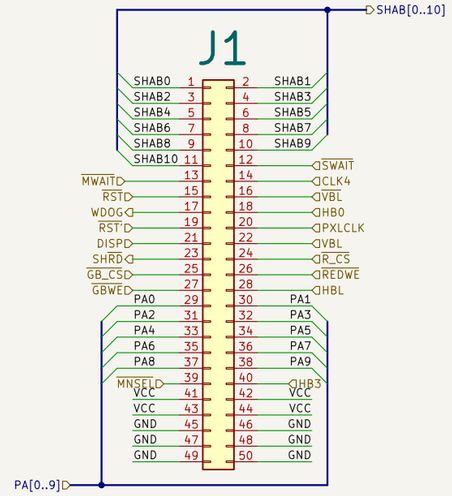
H10



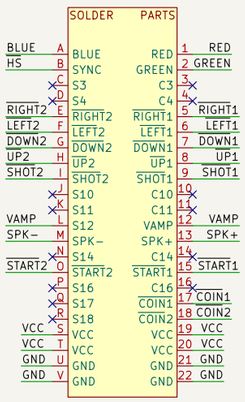
J2



J1



SOLDER PARTS

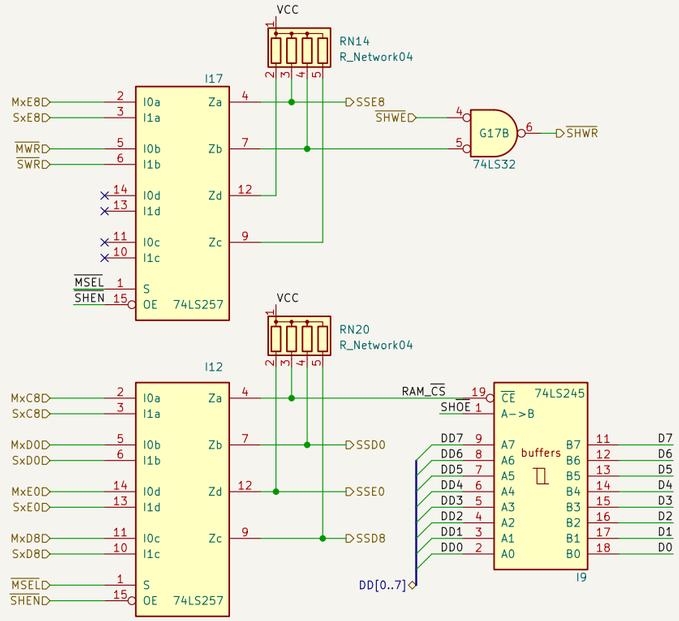


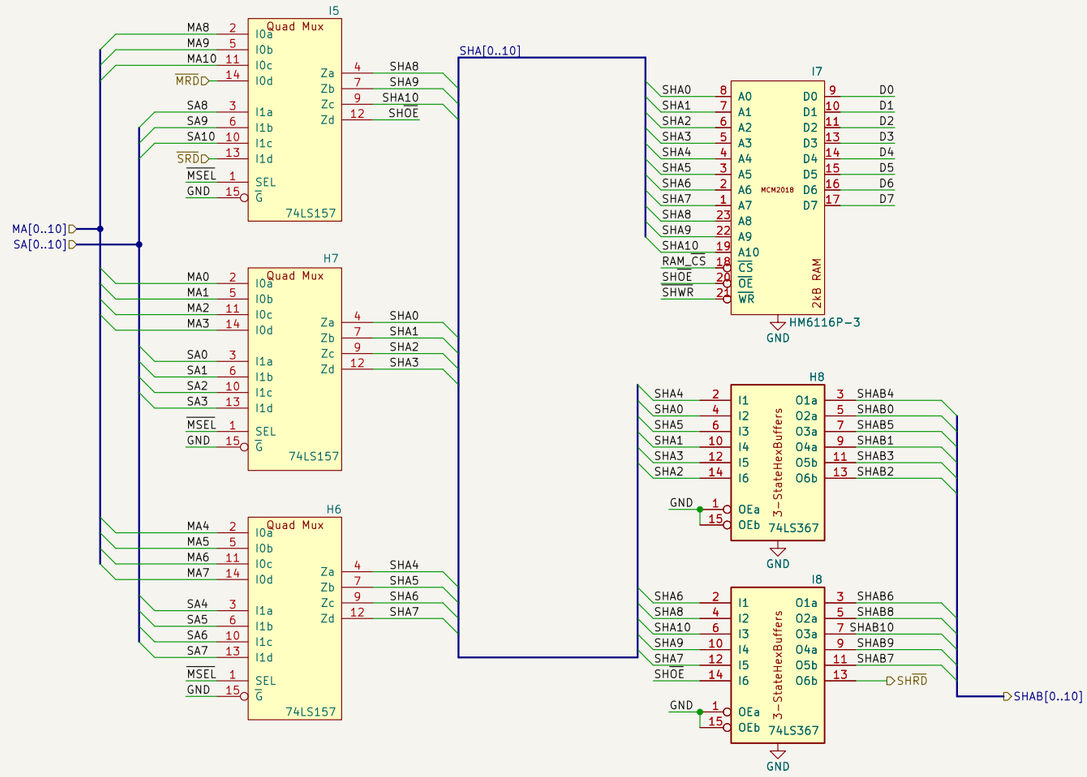
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Sheet: /cpu board/
File: cpu.kicad_sch

Title: CPU board (top board)

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 10.0.4		Id: 9/26





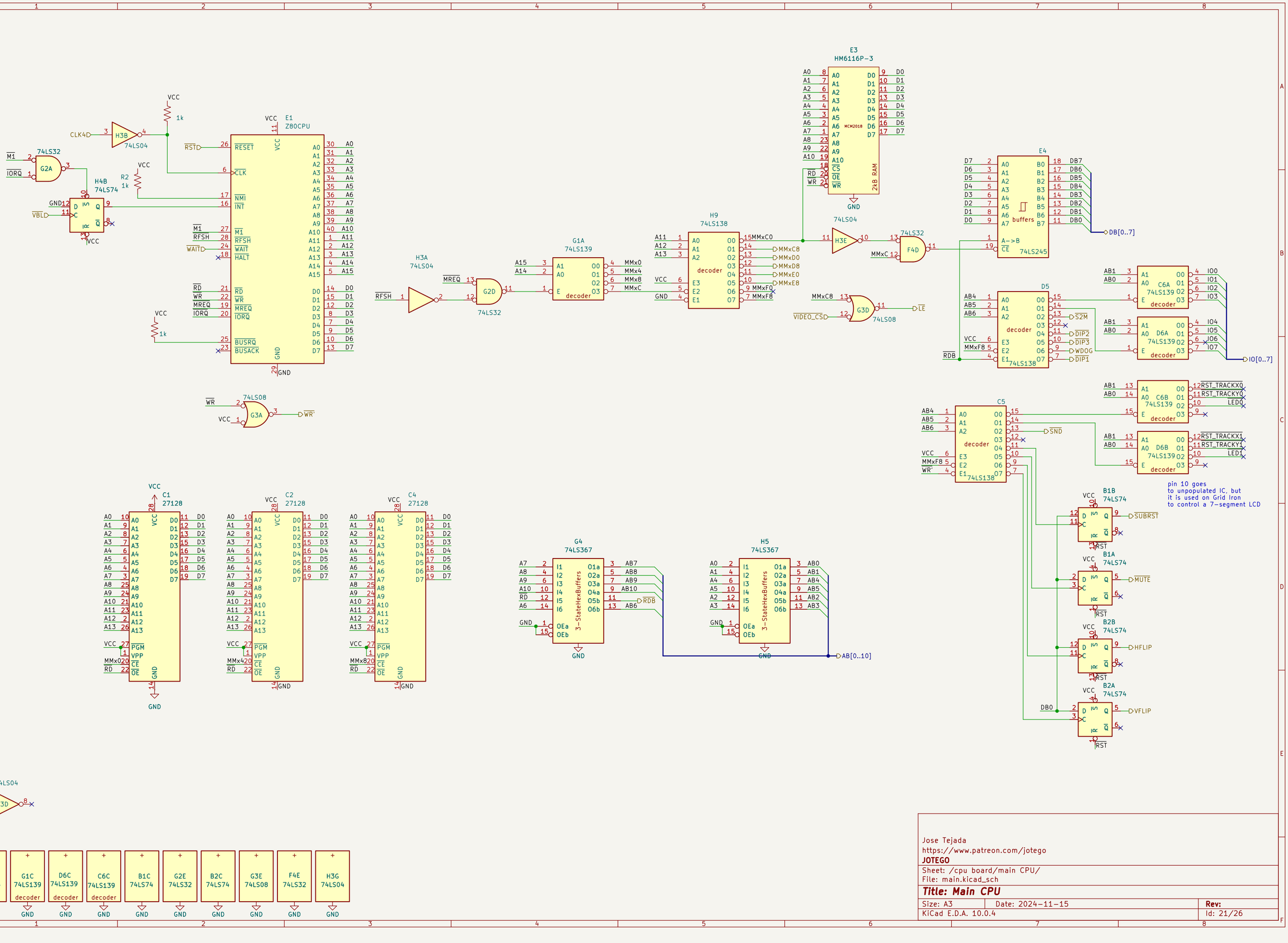


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Sheet: /cpu board/shared/
File: shared.kicad_sch

Title: Shared Memory

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 10.0.4		Id: 10/26


E1
Z80CPU
E3
HM6116P-3
E4
buffers
74LS245
D5
C5
B1B
B1A
B2B
B2A
H9
74LS138
G1A
74LS139
H3A
74LS04
G2A
74LS32
H4B
74LS74
R2
1k
G3A
74LS08
G2D
74LS32
H3E
F4D
G3D
C1
27128
C2
27128
C4
27128
G4
74LS367
H5
74LS367
3-StateHexBuffers
H3D
pin 10 goes
to unpopulated IC, but
it is used on Grid Iron
to control a 7-segment LCD
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Sheet: /cpu board/main CPU/
File: main.kicad_sch
Title: Main CPU
Size: A3
Date: 2024-11-15
KiCad E.D.A. 10.0.4
Rev:
Id: 21/26

Taps to unpopulated sockets

NMICLR

74LS08 E2C

G6B 74LS74

GND SND VCC

M1 IORQ 74LS32 G17C

I16A 74LS74 GND SIRQ VCC

H17A 74LS04 MREQ RFSH 74LS32 G17A

H16B 74LS139 A15 A14 O0 MMx0 O1 MMx4 O2 MMx8 O3 MMx9 E decoder

E17B 74LS32 MMx8 WRB

H16A 74LS139 AB1 AB0 O0 PCMSET O1 PCMCTL O2 GAINSET O3 NMICLR E decoder

VCC G13 Z80CPU

MUTE CPUCLK6 NMI INT M1 RFSH WAIT HALT RD WR MREQ IORQ BUSRQ BUSACK

A0–A15 D0–D7 GND

H13 27128 VCC A0–A14 D0–D7 PGM VPP CE OE GND

I14 HM6116P-3 A0–A10 D0–D7 MMx8 CS RDB OE WRB WR 2kB RAM

D[0..7]

F12 YM2149 VCC D0–D7 DA0–DA7 A8 A9 BDIR BC1 BC2 CLOCK SEL RESET TEST1 VSS GND A B C ASND_A ASND_B ASND_C IOA0–IOA7 CNT0–CNT7 IOB0–IOB7 CNT8–CNT13 MUTE PSGCLK 1k VCC

CNT[0..13]

F15 YM2149 VCC D0–D7 DA0–DA7 A8 A9 BDIR BC1 BC2 CLOCK SEL RESET TEST1 VSS GND A B C BSND_A BSND_B BSND_C IOA0–IOA7 SMP0–SMP7 IOB0–IOB7 SMP8–SMP15 MUTE PSGCLK VCC

SMP[0..15]

74LS04 H17C CLK

I16B 74LS74 VCC D C PSGCLK

74LS04 H17D CPUCLK

MD[0..7]

E11 MD0–MD7 D0–D7 3-state flip flops OE 74LS374 DB0–DB7

E12 DB0–DB7 D0–D7 3-state flip flops OE 74LS374 MD0–MD7

F17 74LS367 3-StateHexBuffers RD A0 A1 WR IORQ OEa OEb GND I1–I6 O1a–O6b RDB ABO AB1 WRB IORQB

F4A 74LS32 MMxC SND

74LS32 G17D WRB MMxC S2M

F11 74LS245 A0–A7 B0–B7 buffers A->B CE RDB DB0–DB7

E10D 74LS08 MMx8 MMxC RDB

D12 74LS174 hex flip flop Cp Mr D0–D5 Q0–Q5 DB1 DB2 DB3 DB0 GAINSET MUTE GAIN1 GAIN2 GAIN3 GAIN0

GAIN[0..3]

PCM PCMCTL PCMSET DBO MUTE SMP[0..15] CNT[0..13] CTL SET DBO MUTE SMP[0..15] A[1..14] PCMSND PCMSND

analog ASND_A ASND_B ASND_C BSND_A BSND_B BSND_C GAIN[0..3] PCMSND ASND_A ASND_B ASND_C BSND_A BSND_B BSND_C GAIN[0..3] PCM SPK+ SPK− SPK+ SPK−

File: analog.kicad_sch

74LS02 F16B WRB 4 ABDIR

74LS02 F16D ABO 13 ABC2

74LS02 F16A RDB 1 ABC1

74LS02 G16B WRB 4 BBDIR

74LS02 G16D ABO 13 BBC2

74LS02 G16A RDB 1 BBC1

IORQB AB1 H17F 74LS04 E17D 74LS32 IOA1 E17C 74LS32 IOA1

BDIR	BC2	BC1	
0	X	0	inactive
1	0	0	latch address
1	1	0	data write
0	0	1	latch address
0	1	1	data read

G6A 74LS74 74LS02 G16C

H17G 74LS04 VCC GND

H16C 74LS139 decoder GND

E10E 74LS08 GND

I16C 74LS74 GND

F16E 74LS02 GND

E2E 74LS08 GND

G16E 74LS02 GND

G6C 74LS74 GND

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Sheet: /cpu board/sound/
File: sound.kicad_sch

Title: Sound

Size: A3	Date: 2024-11-15	Rev:
KiCad E.D.A. 10.0.4		Id: 22/26

SMP[0..15]

E16 74LS191: SMP0 SMP1 SMP2 SMP3 → P0 P1 P2 P3 → A0 A1 A2 A3; SET → PL; VCC → DU; EN → CEN; RC 14 → Cp; RC → RC0; MAXMIN

E15 74LS191: SMP4 SMP5 SMP6 SMP7 → P0 P1 P2 P3 → A4 A5 A6 A7; SET → PL; VCC → DU; EN → CEN; RC0 14 → Cp; RC → RC1; MAXMIN

E14 74LS191: SMP8 SMP9 SMP10 SMP11 → P0 P1 P2 P3 → A8 A9 A10 A11; SET → PL; VCC → DU; EN → CEN; RC1 14 → Cp; RC → RC2; MAXMIN

E13 74LS191: SMP12 SMP13 SMP14 SMP15 → P0 P1 P2 P3 → A12 A13 Q2 Q3; SET → PL; VCC → DU; EN → CEN; RC2 14 → Cp; RC; MAXMIN

JP1: A14 ↔ A13

option to connect
a 32kB ROM (27256)
Used by teedoff

A[1..14]

D14 27128: A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14 → D0 D1 D2 D3 D4 D5 D6 D7; VCC 28; VPP; PGM; CE; OE; GND 14

D16 Quad Mux 74LS157: I0a I0b I0c I0d → Za Zb Zc Zd; I1a I1b I1c I1d; SEL; G ← EN

PULLD: 1k 1k → GND

D17 OKI-M5205 (Semiconductor converter 32kHz): S1 S2 4B/3B D0 D1 D2 D3 T1 T2 RESET XT; DAOUT → PCMSND; VCK; XT; NC NC; GND 12

Y2: 400 P (455 K on bootleg); 101; 471; GND

E17A 74LS32: 1 2 → 3 RC

C11B 74LS74: D ← VCK 11; C ← 10; S ← VCC 14; R ← VCC 14; Q → 9; Q̄ → 8

C11A 74LS74: D ← DB0 2; C ← CTL 3; S ← VCC 14; R ← VCC 14; Q → 5; Q̄ → 6 EN; MUTED 13

E17E 74LS32: VCC GND

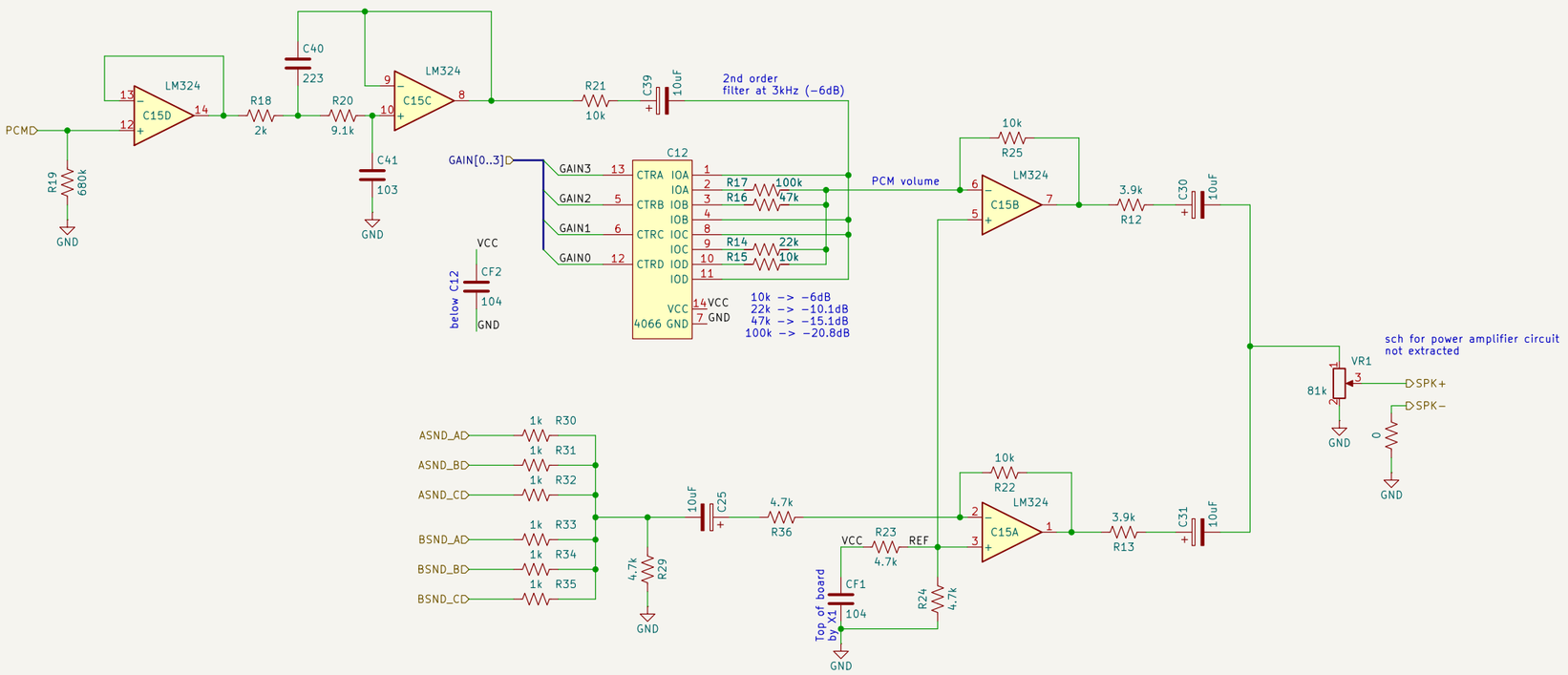
C11C 74LS74: VCC GND

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Sheet: /cpu board/sound/PCM/
File: pcm.kicad_sch

Title: PCM

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 10.0.4		Id: 5/26





Passive component references
when shown, came from original PCB.
The bootleg did not have
references for passives.

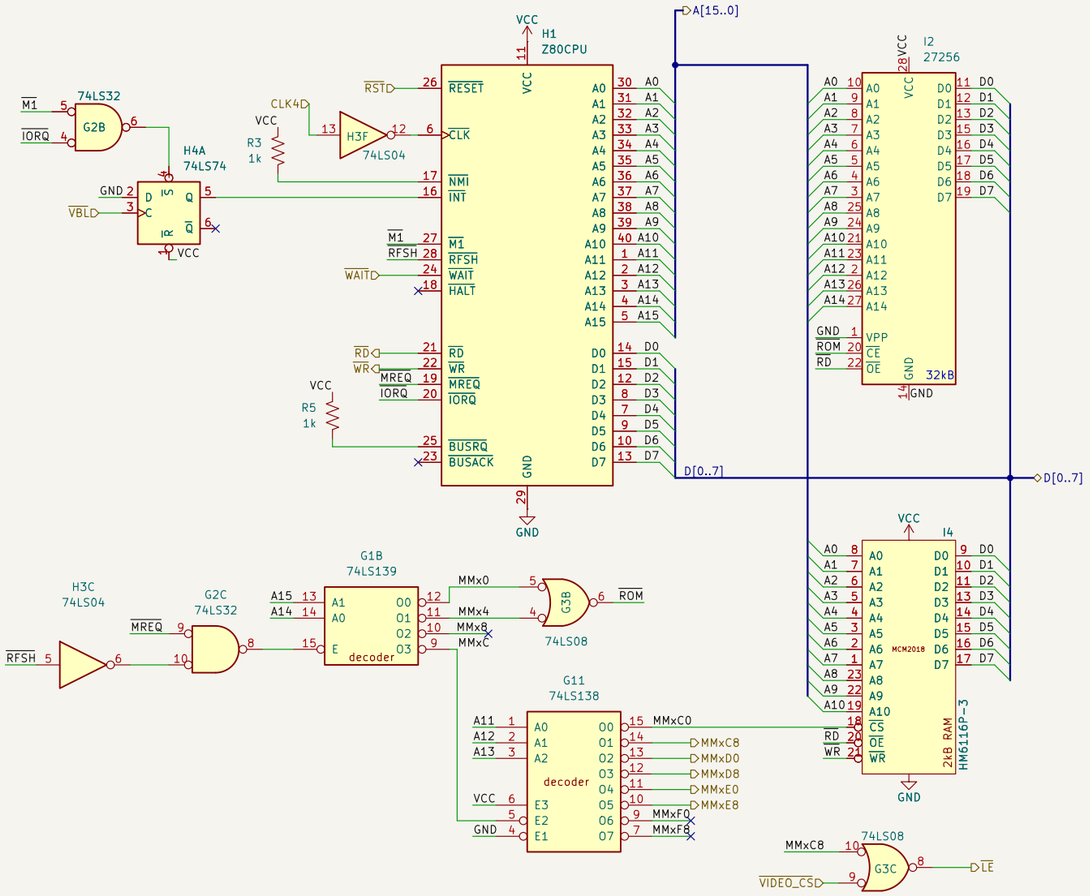
Device components follow bootleg references.
The bootleg counts letters from left to right and numbers from bottom to top
The original counts numbers left to right and letters bottom to top.

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Sheet: /cpu board/sound/analog/
File: analog.kicad_sch

Title: Sound amplifiers

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 10.0.4		Id: 26/26



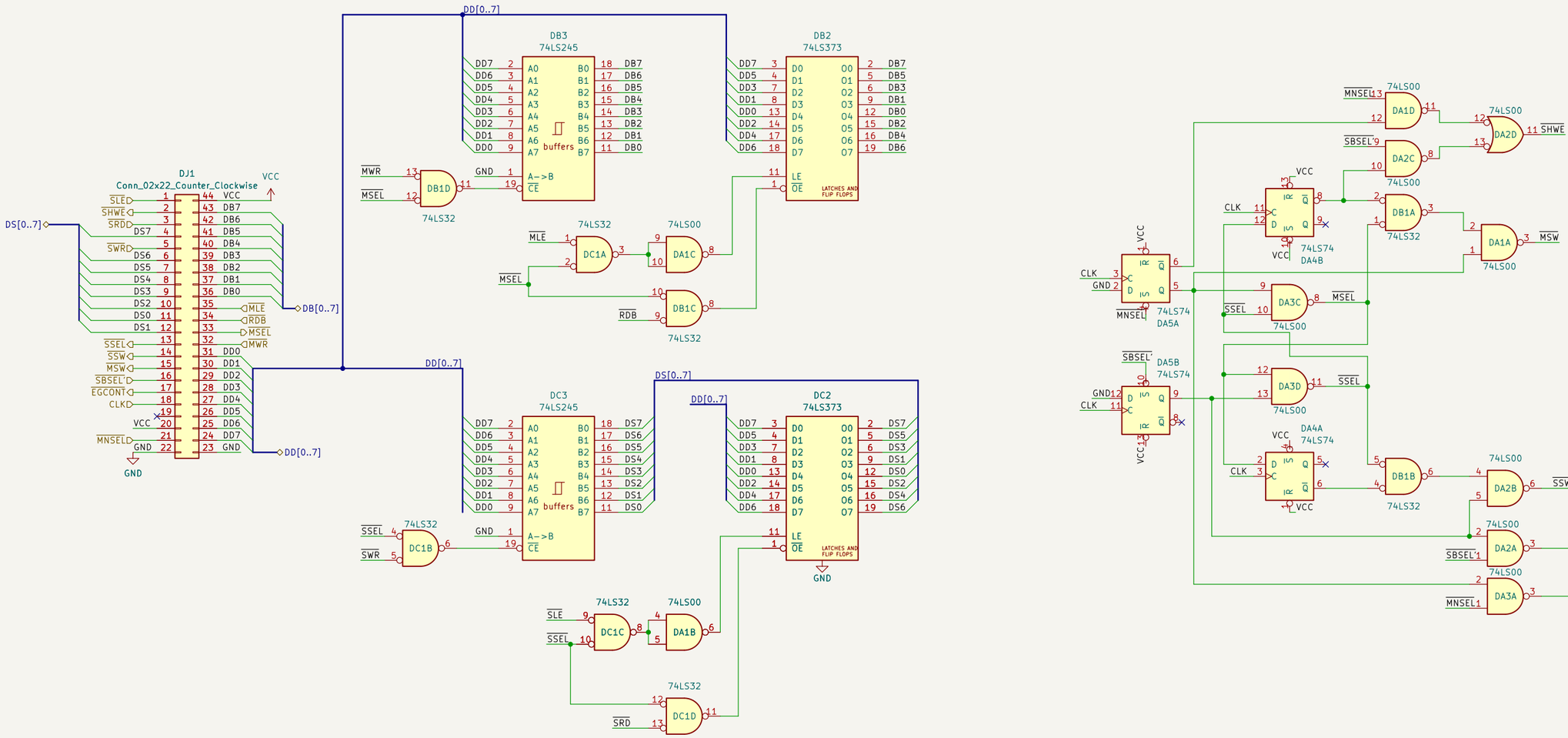
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Sheet: /cpu board/sub CPU/
File: subcpu.kicad_sch

Title: Subordinate CPU

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 10.0.4		Id: 23/26

F5
SW_DIP_x08
E6
74LS244
GND
E5
SW_DIP_x08
F6
74LS244
B3
A2
SW_DIP_x04
3-StateHexBuffers
74LS367
DB[0..7]
VCC
RN17
R_Network04
COIN[1..2]
START[1..2]
B12
Pull-up pin order
may not match the PCB
RN13
RN19
B11
B4
SHOT[1..2]
IO side of the board
contains unpopulated
footprints. The schematics
do not reflect the
full PCB on this side.
RN18
R_Network08_US
DOWN[1..2]
UP[1..2]
RIGHT[1..2]
LEFT[1..2]
maybe through
some pullup?
B5E
74LS04
B5D
B7B
B7C
C10
C9
C8
C7
Quad Mux
74LS157
D10
D9
D8
D7
IO[0..7]
B5G
B7G
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Sheet: /cpu board/IO/
File: io.kicad_sch
Title: Cabinet Input/Output
Size: A3
Date: 2024-11-15
Rev:
KiCad E.D.A. 10.0.4
Id: 24/26



This circuit takes three small boards on the original PCB.
These boards were manufactured by Mitsubishi and contain the circuits seen on the bootleg
- MA7053 x2, 28 pin each
- MA7054 x1, 15 pins
The layout around the daughter board on the bootleg differs slightly from the original in order to adjust the pinout to a single 44-pin board.

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Sheet: /cpu board/daughter/
File: daughter.kicad_sch

Title: Small daughter board

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 10.0.4		Id: 25/26

Resistor networks reference:
RN1 = E9 top
RN2 = below RN1
...
RN11 = above RN12
RN12 = E7 bottom

SYMBOL DOES
NOT MATCH
INTERNAL NETWORK

MSB to common pin 222 Ohm
Each bit up gets 2x (verified)

G and R
are likely
swapped

R from pin 1 to 2 = 2.2kOhm

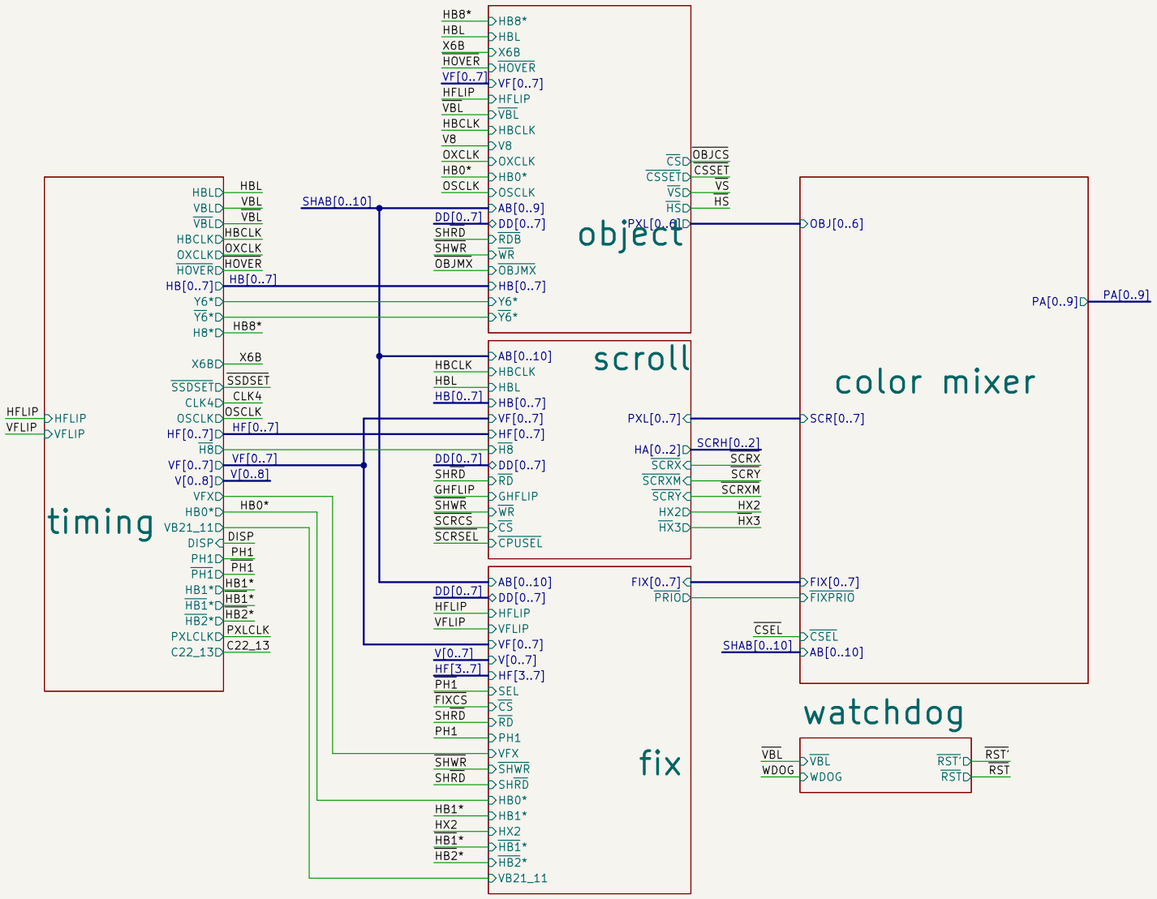
R from pin 1 to 2 = 2.2kOhm

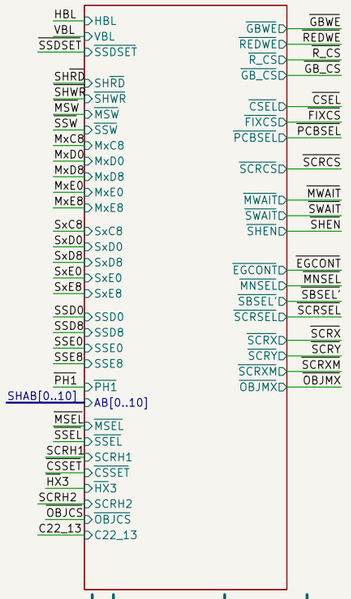
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JOTEGO

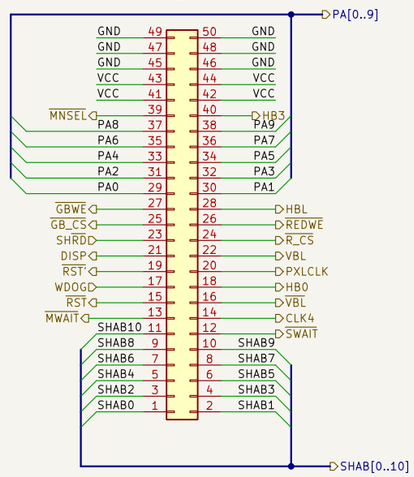
Sheet: /cpu board/palette/
File: palette.kicad_sch

Title: Color Palette

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 10.0.4		Id: #/26

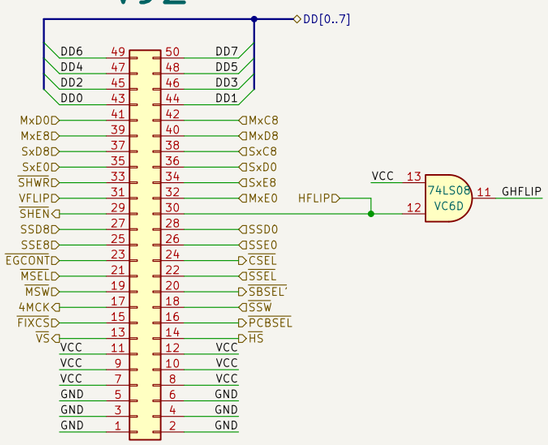






connectors
to CPU board

VJ1 at PCB's top right
VJ2 at PCB's bottom right
pins counted from board bottom up,
pins at the top row of the connector
counted as even

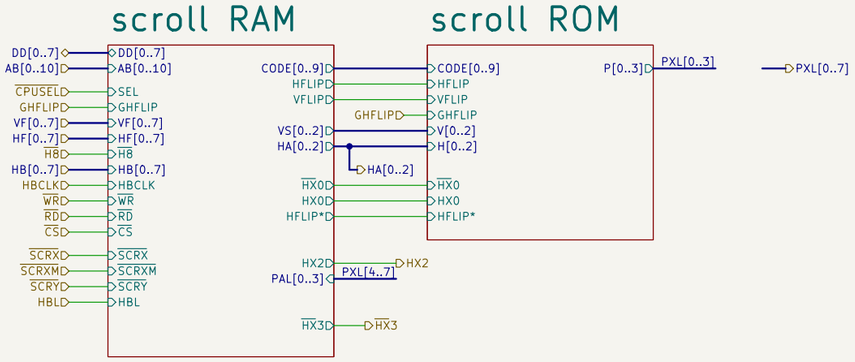


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Sheet: /video board/
File: video.kicad_sch

Title: Video board (bottom board)

Size: A3 | Date: 2024-11-15 | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 11/26

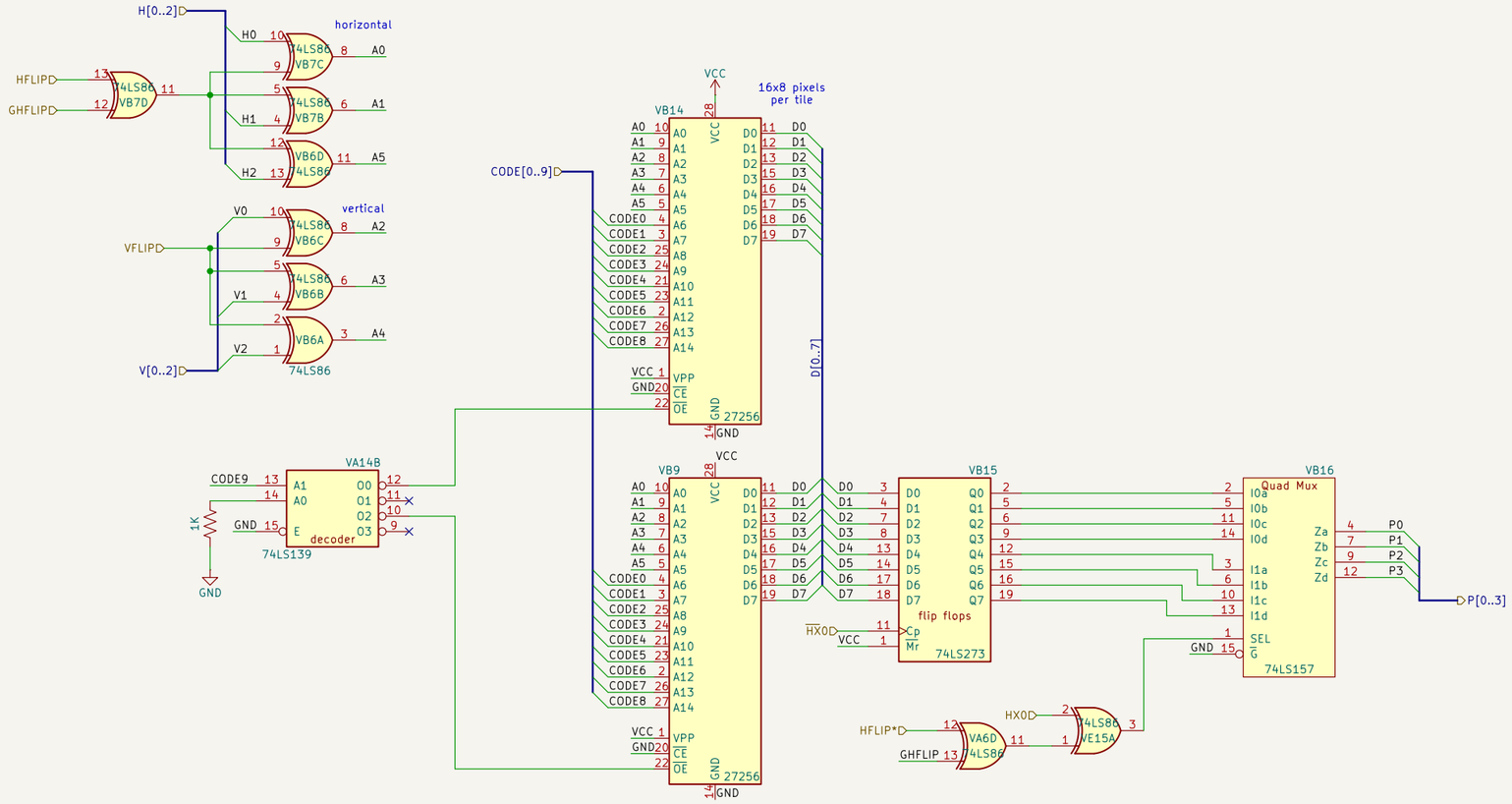


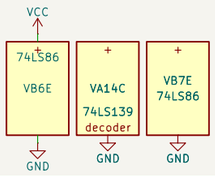
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Sheet: /video board/scroll/
File: scroll.kicad_sch

Title: Scroll Subsystem

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 10.0.4		Id: 12/26



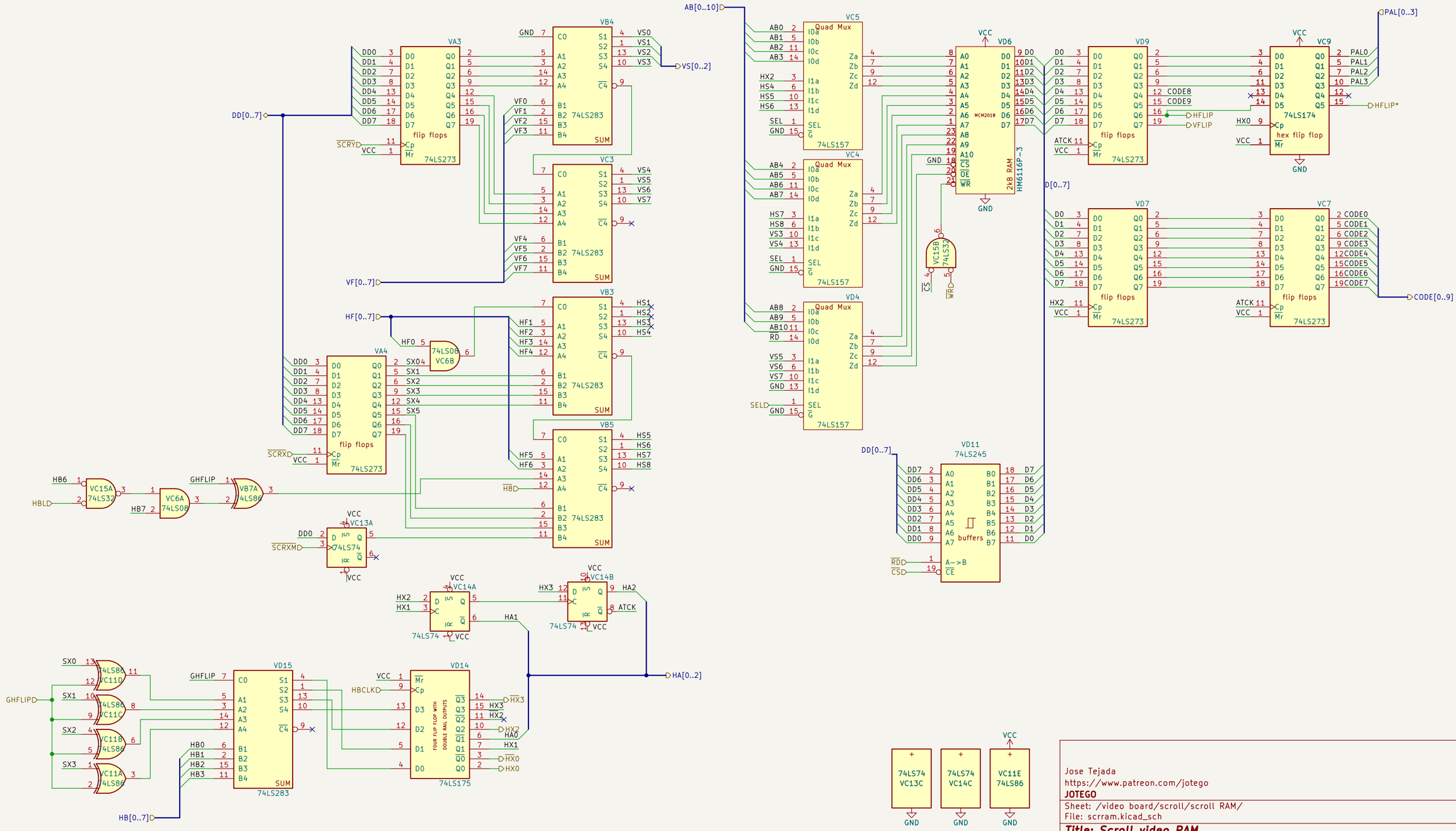


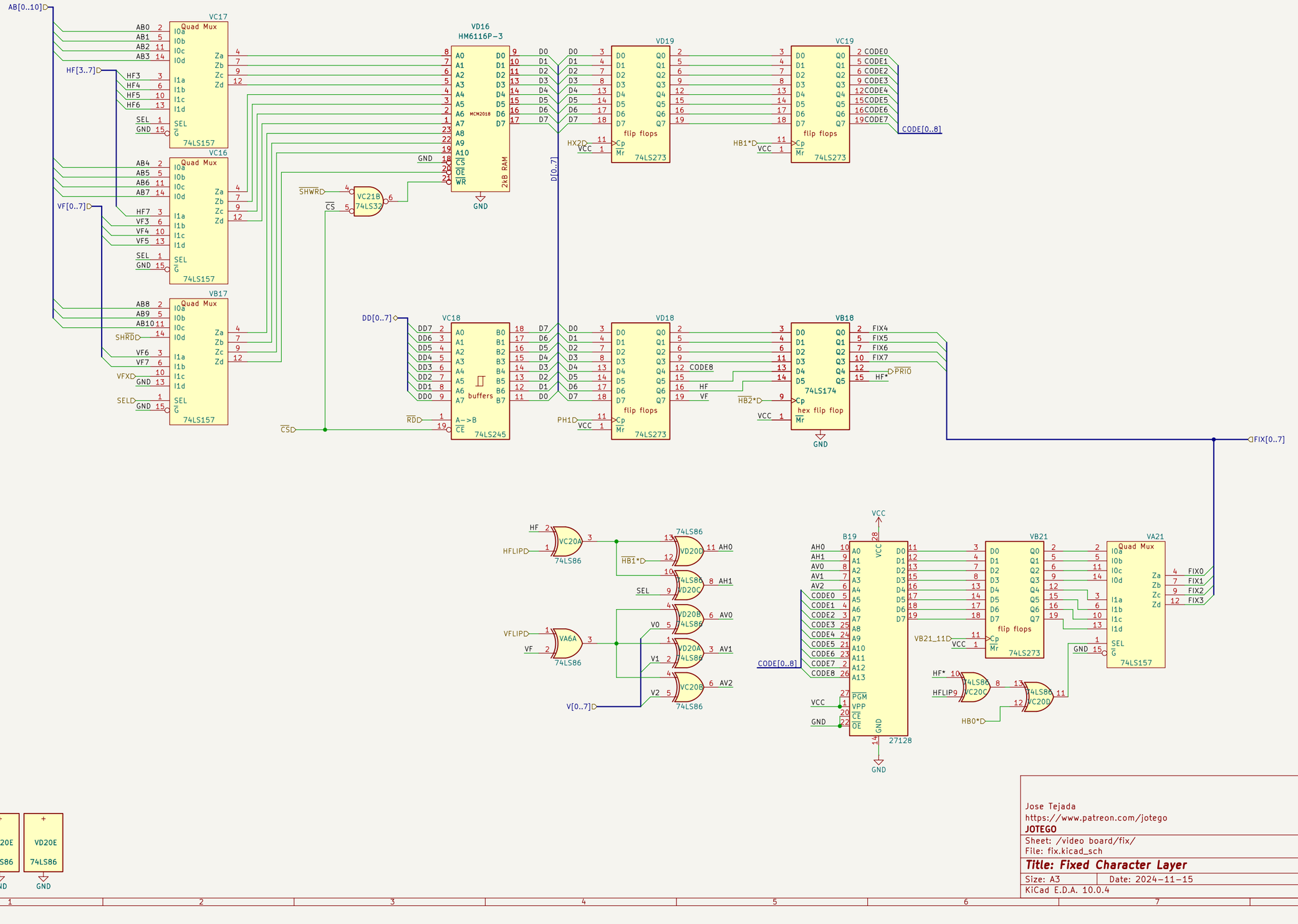
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Sheet: /video board/scroll/scroll ROM/
File: scrrom.kicad_sch

Title: scroll ROM

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 10.0.4		Id: 22/26


AB[0..10]
DD[0..7]
VF[0..7]
HF[0..7]
HB[0..7]
HA[0..2]
VS[0..2]
D[0..7]
CODE[0..9]
PAL[0..3]
HFLIP*
VA3
VB4
VC3
VB3
VA4
VB5
VC5
VC4
VD4
VD6
VD9
VC9
VD7
VC7
VD11
VC13A
VC14A
VC14B
VC15A
VC6A
VB7A
VC6B
VD15
VD14
74LS273
74LS283
74LS157
74LS174
74LS245
74LS74
74LS08
74LS86
74LS32
74LS175
flip flops
hex flip flop
buffers
Quad Mux
MCM2018
HM6116P-3
2kB RAM
SCRYD
SCRXD
SCRXM
HBLD
HFLIP
VFLIP
ATCK
HBCLK
GND
VCC
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JOTEGO
Sheet: /video board/scroll/scroll RAM/
File: scrram.kicad_sch
Title: Scroll video RAM
Size: A3
Date: 2024-11-15
Rev:
KiCad E.D.A. 10.0.4
Id: 23/26

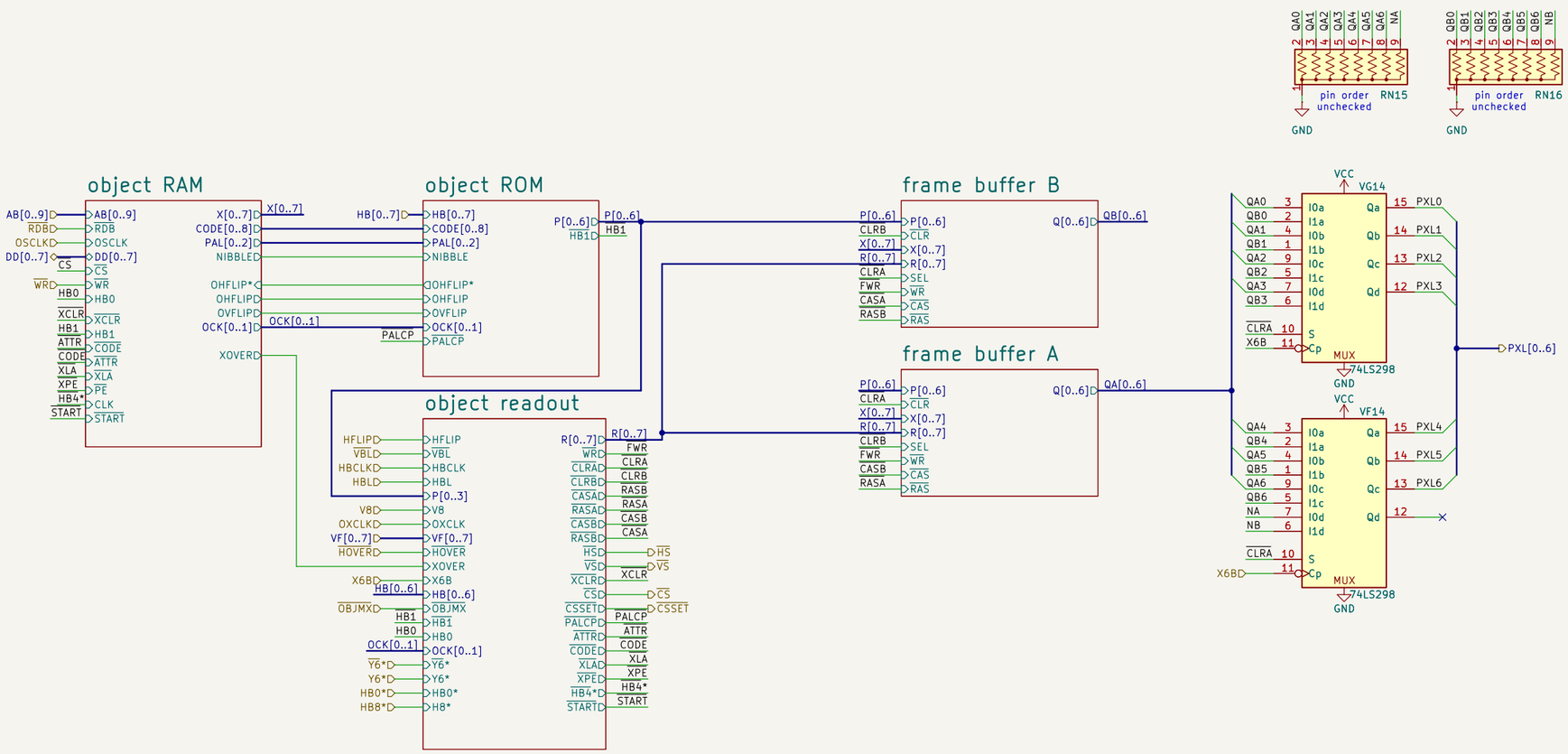


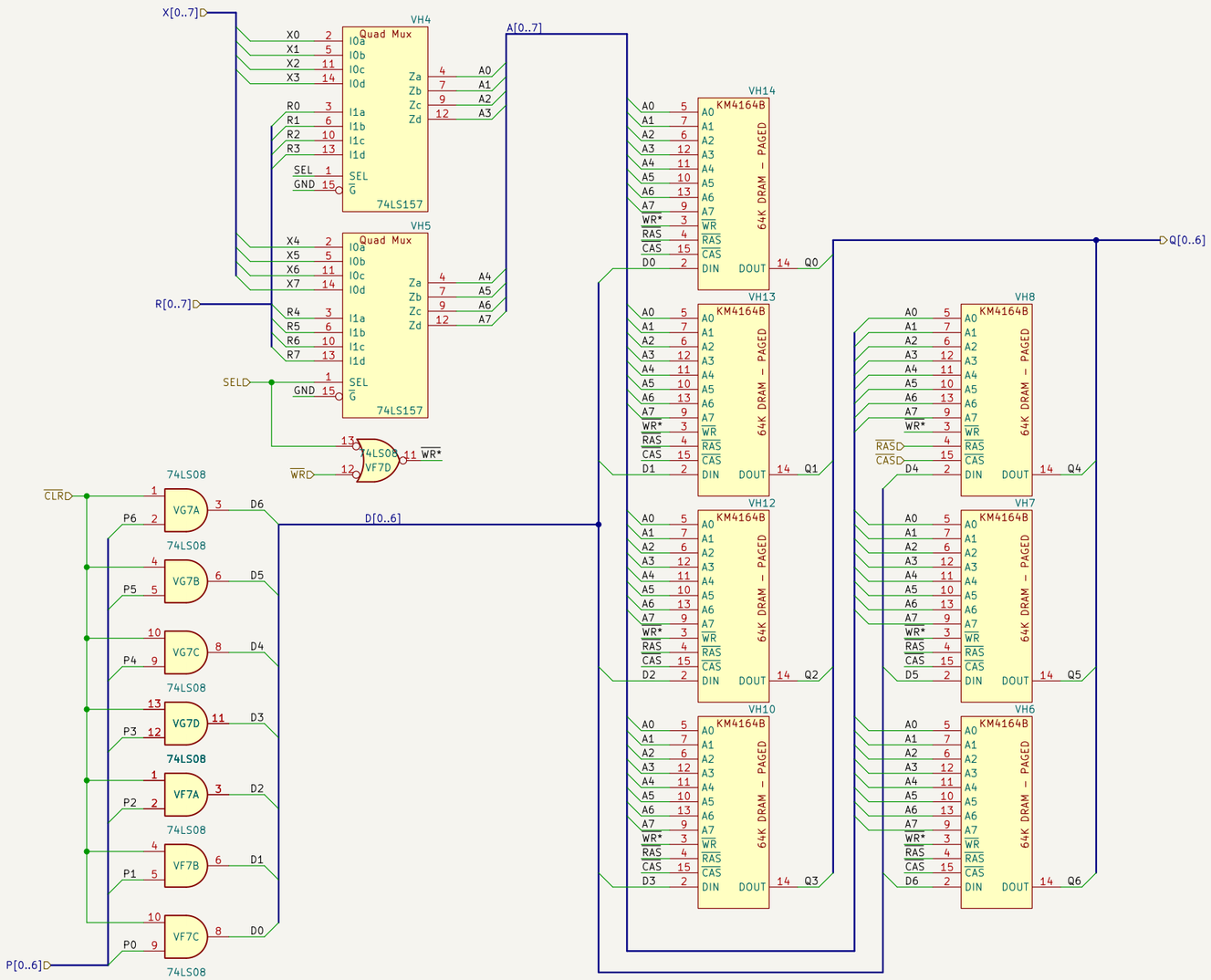
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Sheet: /video board/fix/
File: fix.kicad_sch

Title: Fixed Character Layer

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 10.0.4		Id: 13/26


object RAM
object ROM
frame buffer B
frame buffer A
object readout
pin order unchecked
RN15
RN16
GND
VCC
74LS298
MUX
PXL[0..6]
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Sheet: /video board/object/
File: object.kicad_sch
Title: Object Subsystem
Size: A3
Date: 2024-11-15
KiCad E.D.A. 10.0.4
Rev:
Id: 14/26

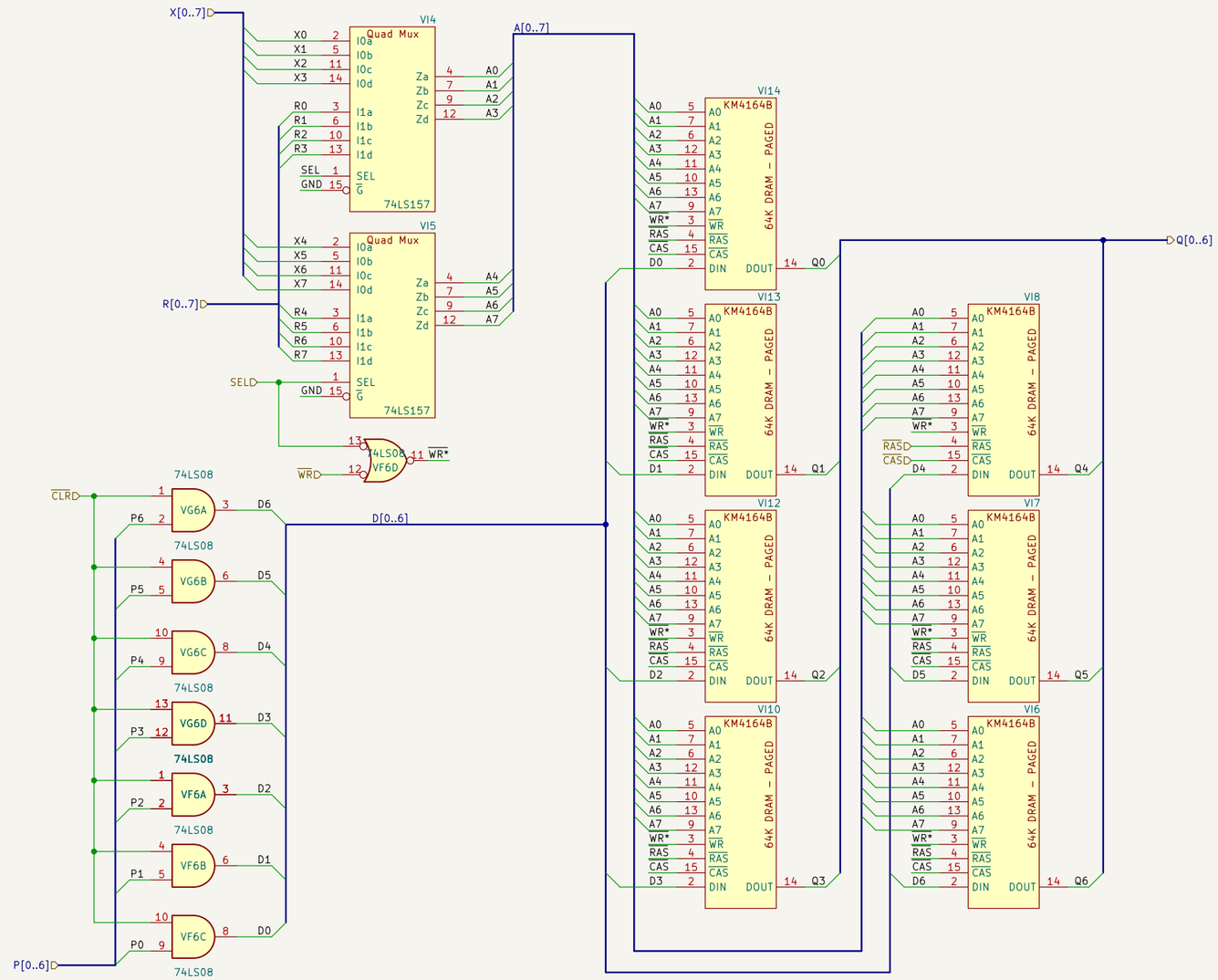


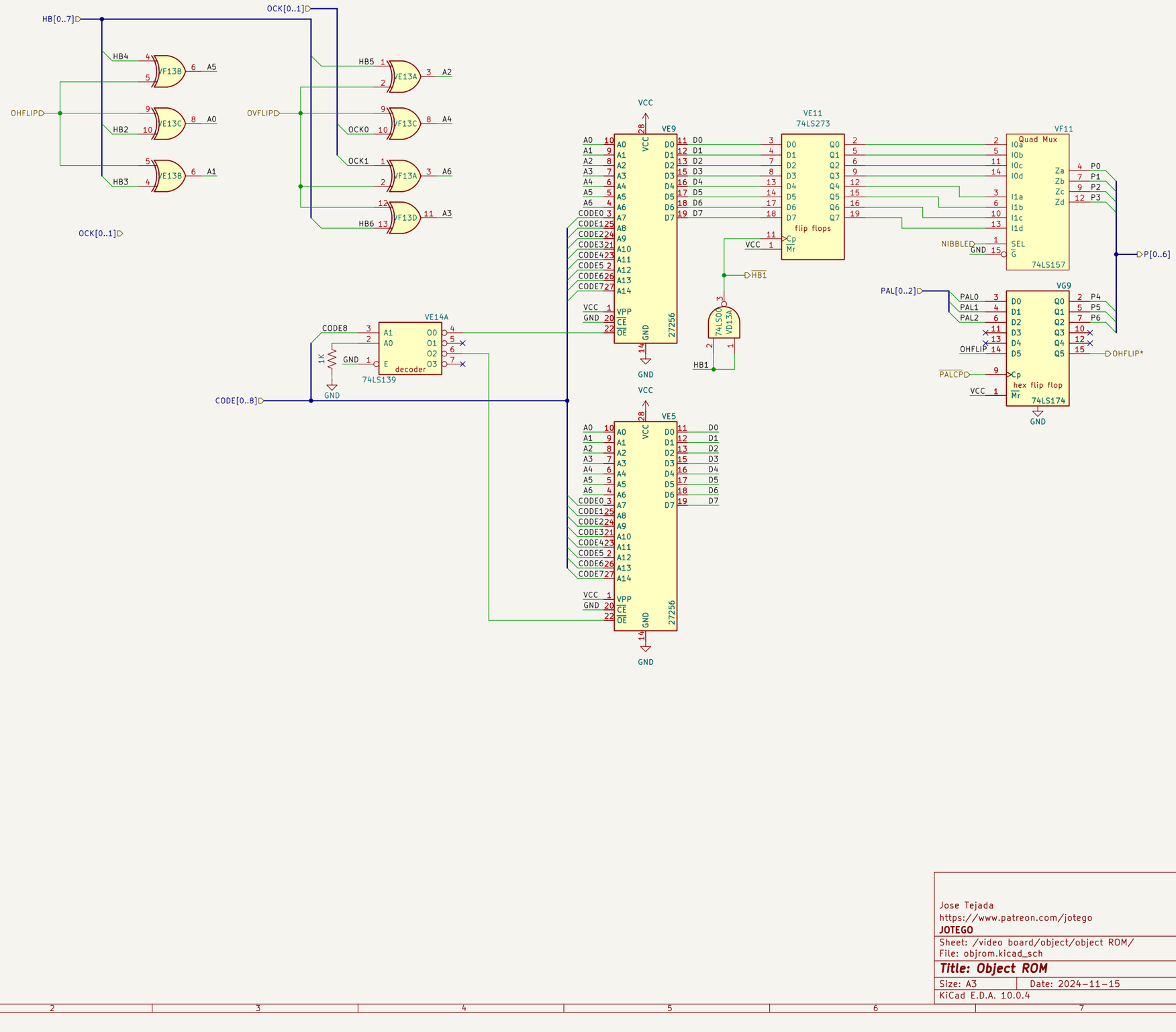
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Sheet: /video board/object/frame buffer B/
File: buffer.kicad_sch

Title: Line Buffer

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 10.0.4		Id: 2/26


X[0..7]
R[0..7]
SELD
WRD
CLRD
P[0..6]
A[0..7]
D[0..6]
Q[0..6]
VI4
Quad Mux
74LS157
VI5
Quad Mux
74LS157
74LS08
VG6A
VG6B
VG6C
VG6D
VF6A
VF6B
VF6C
VG6D
VF6D
VG6E
VF6E
VCC
GND
VI14
VI13
VI12
VI10
VI8
VI7
VI6
KM4164B
64K DRAM - PAGED
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Sheet: /video board/object/frame buffer A/
File: buffer.kicad_sch
Title: Line Buffer
Size: A3
Date: 2024-11-15
Rev:
KiCad E.D.A. 10.0.4
Id: 17/26



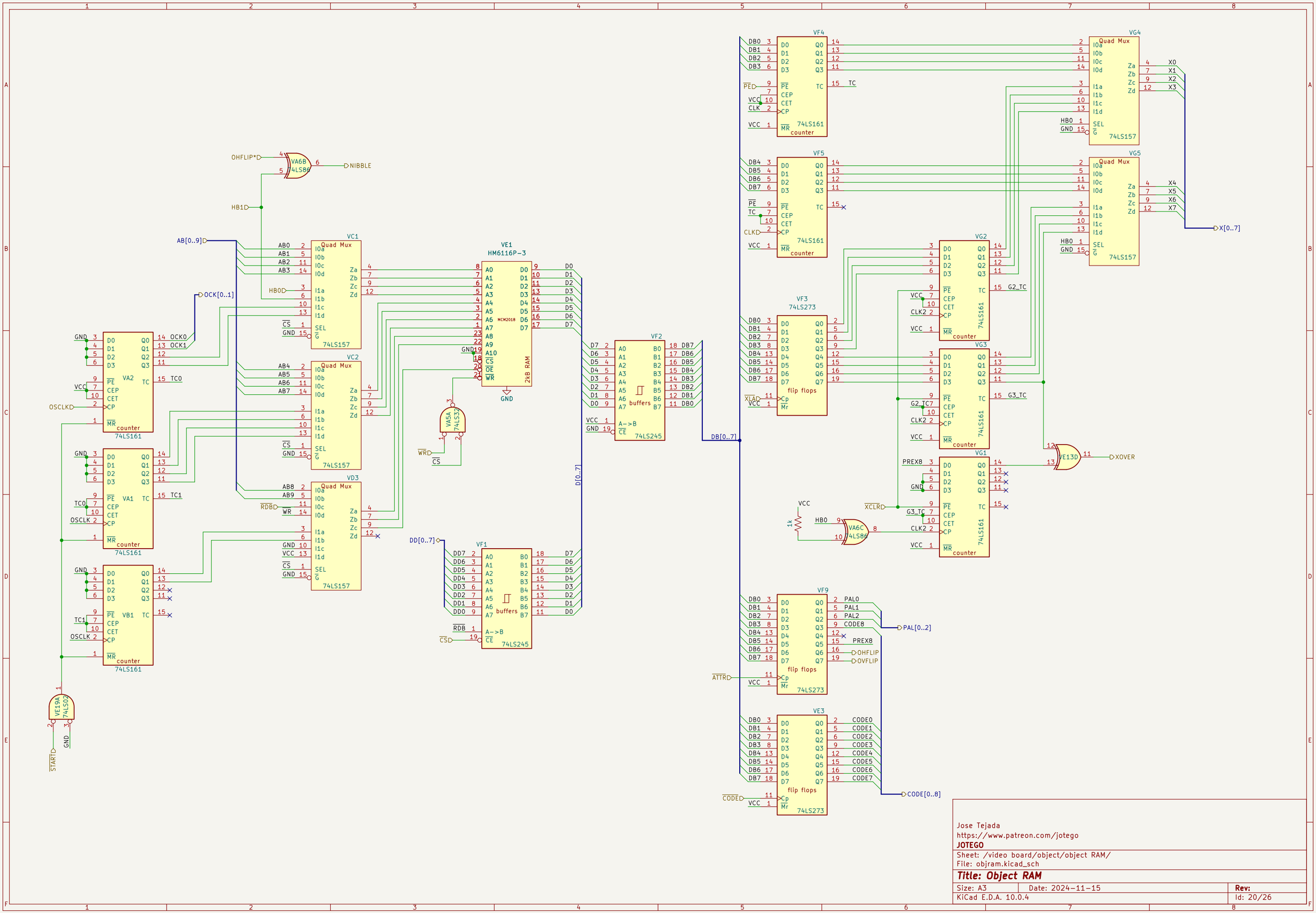
VCC
+
VE14C
74LS139
decoder
GND

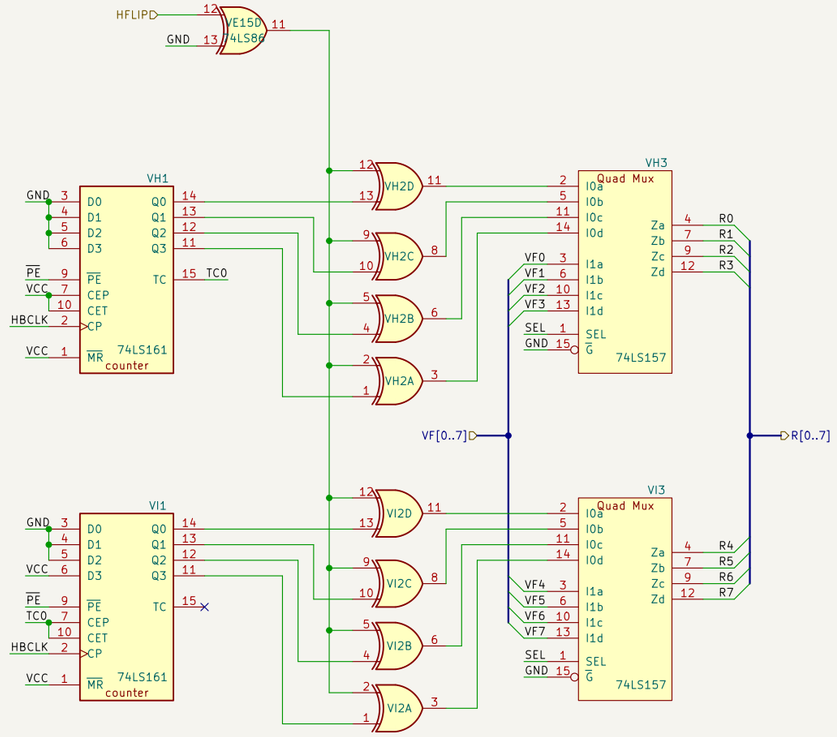
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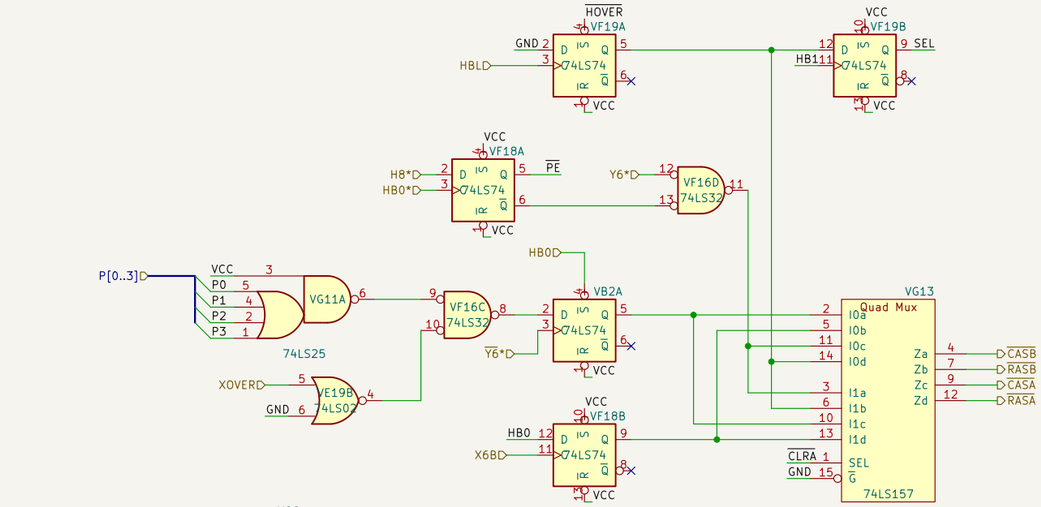
Sheet: /video board/object/object ROM/
File: objrom.kicad_sch

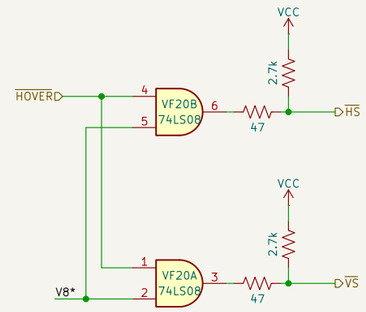
Title: Object ROM

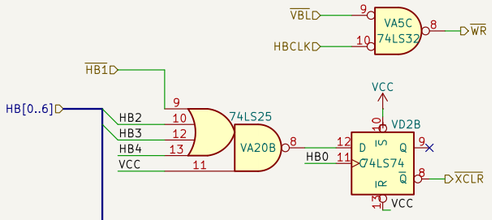
Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 10.0.4		Id: 19/26

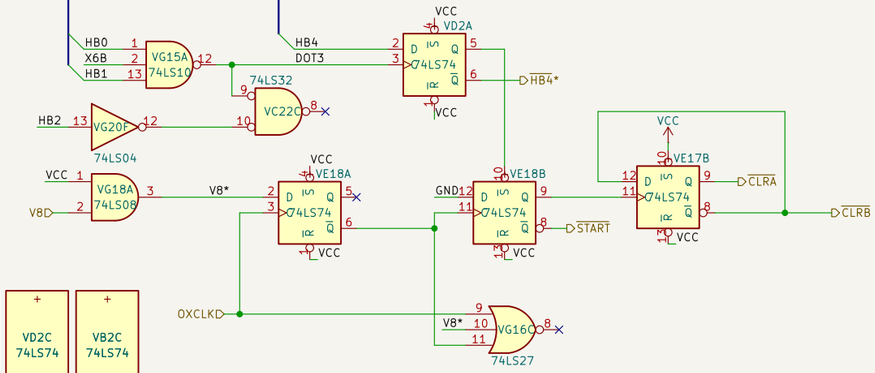

OHFLIP*
VA6B
74LS86
NIBBLE
HB1
AB[0..9]
OCK[0..1]
VC1
Quad Mux
74LS157
VC2
VD3
VA2
VA1
VB1
74LS161
counter
OSCLK
TC0
TC1
VE19A
74LS02
START
VE1
HM6116P-3
2kB RAM
MCM2018
GND
VA5A
74LS32
WR
CS
RDB
VF2
buffers
74LS245
DB[0..7]
DD[0..7]
VF1
D[0..7]
VF4
VF5
VF3
74LS273
flip flops
VG4
VG5
VG2
VG3
VG1
X[0..7]
VE13D
XOVER
XCLR
VA6C
1k
VF9
PAL[0..2]
ATTR
VE3
CODE[0..8]
CODE
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Sheet: /video board/object/object RAM/
File: objram.kicad_sch
Title: Object RAM
Size: A3
Date: 2024-11-15
Rev:
KiCad E.D.A. 10.0.4
Id: 20/26

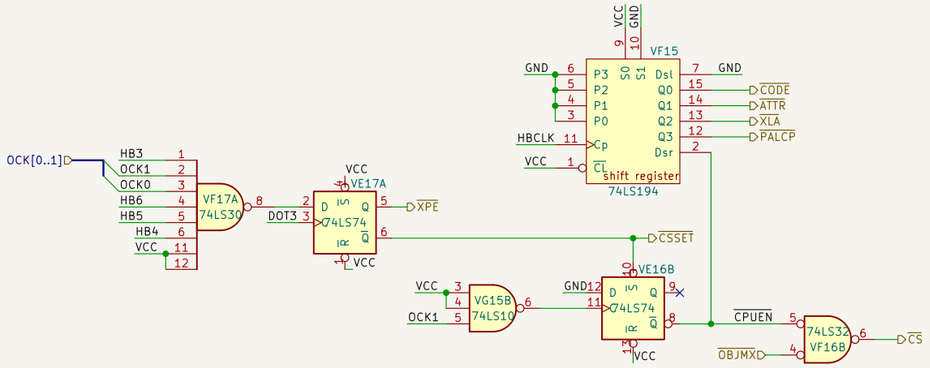


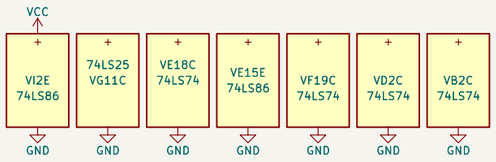












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Sheet: /video board/object/object readout/
File: objread.kicad_sch

Title: Object readout sequencer

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 10.0.4		Id: 21/26

(3.57MHz on bootleg)

Y3
4.0MHz
101
101
GND

VI22A 4069
VI22B 4069
VH22A 74LS04
CLK4

clock divider by 3

VI20B 74LS107
VI20A 74LS107
Y6
X6
HCLK
6.144 MHz
VCC

K15
330
VJ15B
74LS04
C3
101
330
VJ15A
74LS04
Y1
18.432
(18.000 on bootleg)

VI16 74LS161
VI15 74LS161
V[0..8]
V0
VTC
VCLK
LSB
MSB
counter
VG20F
74LS04

VI18 74LS161
VI19 74LS161
VG19A 74LS74
HCLK
H0
3MHz
TCH
H1
H2
H3
H4
H5
H6
H7
H8

VG20C
VG21A
HOVER
VG19B
VBL
VBL*

74LS00
VH20B
74LS74

74LS86
VH16D
VH16C
VH16B
VH16A
VF0
VF1
VF2
VF3
V1
V2
V3
V4
V5
V6
V7

VH15D
VH15C
VH15B
VH15A
VF4
VF5
VF6
VF7

VG18B
VFLIP
VF[0..7]

74LS20
VG17B
VG18C
VG17A
VG20D
X6B
TCH
HB6
HB7
HB5
VFX
HB1*
object scan clock
OSCLK
HBL
74LS27
VG16A
VG20A
OXCLK
HBCLK
HB0*
VG16B
VE21B
SSDSET

VH17
VI17
3-StateHexBuffers
74LS367
HB[0..7]
HB0
HB1
HB2
HB3
HB4
HB5
HB6
HB7
X6B
HBCLK
Y6*

VG18D
HFLIP
HF[0..7]
HF0
HF1
HF2
HF3
HF4
HF5
HF6
HF7
VH18D
VH18C
VH18B
VH18A
VH19A
VH19B
VH19C
VH19D

VC22B
VD21A
C22_13

VF21
74LS175
HB0*
VB21_11
HB1*
VFX
HB2*

VE20C
PXLCLK

VF20C
VH20C
VG21B
VF22A
VG22A
VG22B
VH22E
VF20D
PH1
PH1*
H8*
DISP

VE16C 74LS74
VG19C 74LS74
VJ15G 74LS04
VG18E 74LS08
VG15D 74LS10
VH20E 74LS00
VH18E 74LS86
VH19E 74LS86
VH16E 74LS86
VI20C 74LS107
VF22C 74LS74
VG22C 74LS74
VG20G 74LS04
VG21C 74LS74
VH22G 74LS04
VG17C 74LS20
VG16D 74LS27
VF20E 74LS08
VE20E 74LS32
VE21G 74LS04

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Sheet: /video board/timing/
File: timing.kicad_sch

Title: Video Timing

Size: A3 | Date: 2024-11-15 | Rev:
KiCad E.D.A. 10.0.4 | Id: 15/26

FIX[0..7]
OBJ[0..6]
SCR[0..7]
AB[0..10]

VE21D 9 8
74LS32
VE22C 9 8
OBJOPA 10
VD22A 1 2 3 S0
74LS08
CSEL

S1 S0 SEL
0 0 CPU
0 1 FIX
1 0 OBJ
1 1 SCR

FIXPRIO 5 VE21C 6
VCC 11
OBJ3 13
OBJ2 12
OBJ1 10
OBJ0 9
VG11B 8
74LS25
OBJOPA 13
74LS32
VE22D 12 11
FIX3 1
FIX2 2
FIX1 4
FIX0 5
74LS25
VA20A 3 6
OBJ has priority over FIX
FIXOPA
74LS08
VD22B 5 4 6 S1

VA19
AB9 6 GND 5 VCC 4 GND 3 I0a I1a I2a I3a
GND 1 Ea
AB10 10 GND 11 12 VCC 13 I0b I1b I2b I3b
GND 15 Eb
S0 14 S1 2 S0 S1
Mux 4-1
74LS153
Za 7 PA8
Zb 9 PA9

VA18
AB7 6 FIX6 5 OBJ6 4 SCR6 3 I0a I1a I2a I3a
GND 1 Ea
AB8 10 FIX7 11 GND 12 SCR7 13 I0b I1b I2b I3b
GND 15 Eb
S0 14 S1 2 S0 S1
Mux 4-1
74LS153
Za 7 PA6
Zb 9 PA7

VA17
AB5 6 FIX4 5 OBJ4 4 SCR4 3 I0a I1a I2a I3a
GND 1 Ea
AB6 10 FIX5 11 OBJ5 12 SCR5 13 I0b I1b I2b I3b
GND 15 Eb
S0 14 S1 2 S0 S1
Mux 4-1
74LS153
Za 7 PA4
Zb 9 PA5

PA[0..9]

VA16
AB3 6 FIX2 5 OBJ2 4 SCR2 3 I0a I1a I2a I3a
GND 1 Ea
AB4 10 FIX3 11 OBJ3 12 SCR3 13 I0b I1b I2b I3b
GND 15 Eb
S0 14 S1 2 S0 S1
Mux 4-1
74LS153
Za 7 PA2
Zb 9 PA3

VA15
AB1 6 FIX0 5 OBJ0 4 SCR0 3 I0a I1a I2a I3a
GND 1 Ea
AB2 10 FIX1 11 OBJ1 12 SCR1 13 I0b I1b I2b I3b
GND 15 Eb
S0 14 S1 2 S0 S1
Mux 4-1
74LS153
Za 7 PA0
Zb 9 PA1

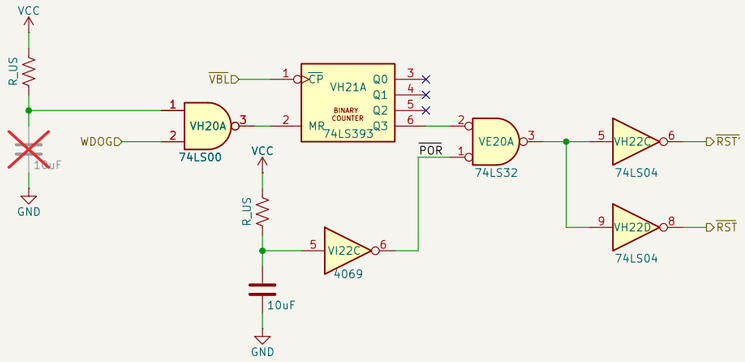
VCC
VE22E 74LS32 GND
VF18C 74LS74 GND
VD21C 74LS74 GND
VF16E 74LS32 GND
VC22E 74LS32 GND
VH15E 74LS86 GND
VA20C 74LS25 GND
VD22E 74LS08 GND
VE13E 74LS86 GND
VF13E 74LS86 GND
VH2E 74LS86 GND
VA5E 74LS32 GND
VE17C 74LS74 GND
VF17B 74LS30 GND

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Sheet: /video board/color mixer/
File: colmix.kicad_sch

Title: Color mixer

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 10.0.4		Id: 16/26



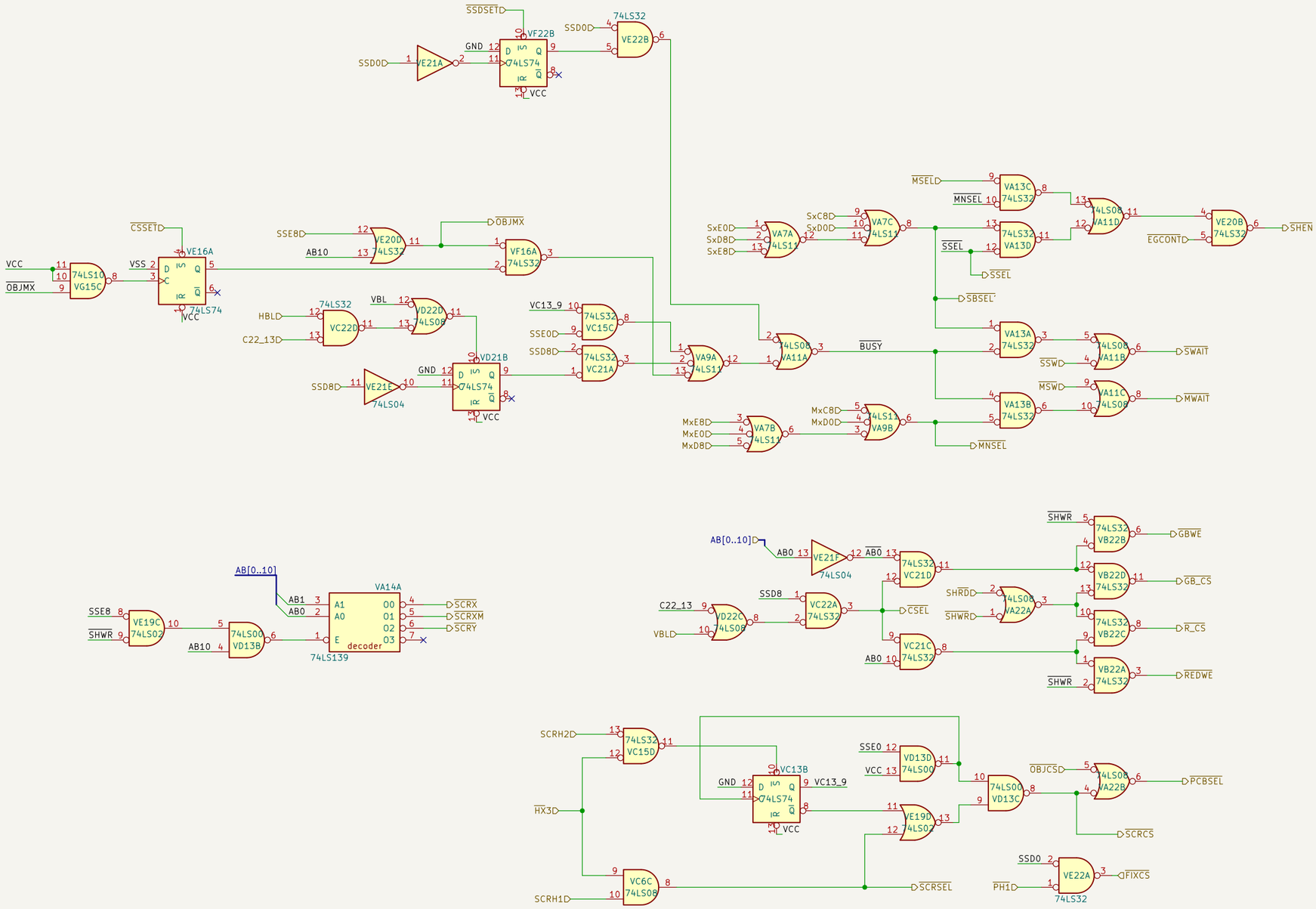


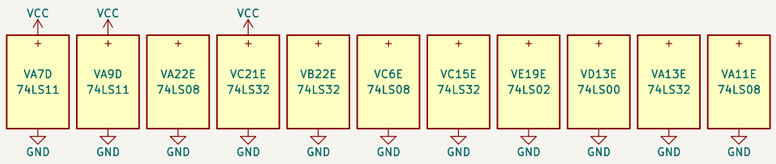
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Sheet: /video board/watchdog/
File: watchdog.kicad_sch

Title: Watchdog Circuit

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 10.0.4		Id: 24/26





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Sheet: /video board/address decoder/
File: decoder.kicad_sch

Title: Address decoder

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 10.0.4		Id: 25/26